

Port Operations Copilot

AI-Driven Congestion Hotspot Prediction, Prescriptive Google OR-Tools Optimization, & 72-Hour Rolling Operational Planning

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1. The Problem: The \$10B+ Global Port Congestion Crisis

The 2021 LA / Long Beach port crisis left 100+ container vessels idling offshore for weeks.

- **Manual Spreadsheet Scheduling:** Berths, cranes, and yard slots are allocated via static spreadsheets, blind to holistic constraints.
- **Reactive Bottleneck Discovery:** Congestion is detected only after ships have already formed long offshore queues.
- **Delayed Rerouting Decisions:** Shippers and terminal operators lack foresight to divert incoming calls to under-utilized partner berths before bottlenecks escalate.
- **Multi-Billion Dollar Impact:** Demurrage fees, intermodal rail disruption, and massive supply chain inflation.

2. The Solution: Intelligent Port Operations Copilot

A closed-loop AI & Mathematical Optimization decision support system.

- **Predictive Risk Engine:** Sliding-window ML (Random Forest) + domain heuristics forecast congestion hotspots up to 72h ahead.
- **Google OR-Tools CP-SAT:** Exact mathematical constraint programming schedules non-overlapping berth windows and eliminates waiting times.
- **Dynamic Crane Allocation:** Allocates 2-5 STS cranes per vessel based on TEU workload and vessel priority class.
- **Proactive Alternate Routing:** Recommends feasible diversions with verified draft/LOA clearance before vessels arrive.

3. IBM Bob & watsonx AI Integration

IBM Bob is deeply load-bearing across development and runtime operations.

- **Engineering Acceleration:** IBM Bob steered architecture design, OR-Tools CP-SAT modeling, and automated test synthesis.
- **Conversational Copilot:** Grounded in live port state (vessel ETAs, berth schedules, crane metrics, risk scores).
- **watsonx Granite 3.0 Integration:** Orchestrates domain-specific prompts with live telemetry injection.
- **6-Point Explainability Audit:** Every assignment answers What happened, Why problem, System prediction, Recommended action, Why selected, and Constraints verified.

4. Measurable Operational Impact

Comparing Baseline (Naive FIFO) vs AI & OR-Tools Optimized Schedule:

Operational Metric	Unoptimized Baseline	AI & OR-Tools Optimized	Impact Delta
Average Vessel Waiting Time	41.1 Hours	28.5 Hours	30.7% Reduction
Maximum Vessel Queue Delay	107.5 Hours	68.2 Hours	36.6% Reduction
Congested Delayed Vessels (>3h)	32 Vessels	18 Vessels	14 Vessels Saved
Fleet Port Dwell Time Saved	—	—	504+ Vessel-Hours Saved
Proactive Reroutings Executed	0 (Reactive)	3 Diversions	Hotspot Eradicated

5. Future Vision & Enterprise Roadmap

From Hackathon Prototype to Global Maritime TOS Standard:

- **Real-Time AIS Vessel Telemetry:** Live satellite feed ingestion for sub-hourly ETA updates.
- **Intermodal Rail & Drayage Truck Sync:** Extending optimization into container yards and railheads.
- **Port Multi-Terminal Federation:** Unified optimization across entire port authorities (e.g. LA, Long Beach, Rotterdam, Singapore).
- **watsonx Governance:** Full model lifecycle monitoring, compliance tracking, and audit logging.